

SHIVAM KUMAR

Java Backend Developer

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PROFESSIONAL SUMMARY

Java Backend Developer and final-year B.Tech CSE student (Class of 2027) building secure, distributed backends and RESTful microservices with Java 17, Spring Boot 3, Spring Security, Hibernate ORM, and Spring Data JPA. Architected stateless JWT authentication, Role-Based Access Control (RBAC), multi-stage Docker containerization, and database persistence optimization with MySQL and Redis caching. 2x Tech Hackathon Finalist, designing and deploying production-ready full-stack prototypes within competitive timeframes.

TECHNICAL SKILLS

Languages: Java (Java 8/17/21 - Collections Framework, Streams API, Multithreading, Concurrency, Lambda Expressions, OOP), SQL, JavaScript (ES6+), Python (Basic), C

Backend & Frameworks: Spring Boot 3.4+, Spring Security (JWT, OAuth2, BCrypt), Spring Data JPA, Hibernate ORM, Spring MVC, Microservices Architecture, RESTful API Design, WebSocket, Thymeleaf

Databases & Caching: MySQL 8, PostgreSQL, Redis (In-Memory Caching & Session Storage), MongoDB, H2 Database

DevOps, Cloud & Tools: Docker, Docker Compose, Git, GitHub, Maven, Postman, Cloudinary CDN, Render (IaC Blueprint), JUnit 5, Mockito, IntelliJ IDEA, VS Code

Core Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), System Design, MVC Architecture, Database Normalization, ACID Transactions, Agile/Scrum

PROFESSIONAL EXPERIENCE

Backend Engineering Intern - Analyze InfoTech, Lucknow

May 2026 - Jul 2026

- Engineered high-performance RESTful APIs for an e-commerce platform supporting 2 distinct user tiers (Admin, Customer), reducing payload latency by 25%.
- Secured 20+ user endpoints by architecting stateless JWT authentication with BCrypt encryption, eliminating unauthorized API access vulnerabilities.
- Structured normalized relational tables in MySQL and document collections in MongoDB for 1,000+ catalog items, improving query execution speed by 35% via targeted database indexing.

PROJECTS

Tool & Equipment Rental System (ToolShare) | Java 17, Spring Boot 3.4, JWT, JPA, MySQL, Docker, React 18

[Repository](#)

- Architected a full-stack equipment rental marketplace backend with Java 17 and Spring Boot 3.4, managing 15+ REST endpoints with geolocation search.
- Programmed strict Role-Based Access Control (ROLE_USER, ROLE_OWNER) using Security 6 filters and stateless JWTs.
- Automated a 4-stage rental booking lifecycle (PENDING to COMPLETED) featuring conflict-locking algorithms that processed 50+ concurrent requests with 0% double-booking error rate.
- Containerized the full application stack using multi-stage Docker builds and Docker Compose, reducing production image size by 40% and automating Render cloud deployment.

Electronic Voting Machine (EVM) System | Java 17, Spring Boot, Hibernate ORM, MySQL, Thymeleaf

[Repository](#)

- Built a digital voting portal using MVC architecture and Hibernate ORM, handling 500+ simulated concurrent voters with 99.9% application uptime.
- Configured RBAC security filters separating 2 user roles (Admins and Voters), accelerating candidate verification processing time by 45%.
- Enforced atomic database transactions to eliminate duplicate voting attempts, guaranteeing 100% ballot immutability and data integrity across 1,000+ cast votes.
- Optimized database throughput using Data JPA indexed queries and custom repository methods, cutting read latency by 30% during multi-round mock elections.

CERTIFICATIONS & ACHIEVEMENTS

- 2x Tech Hackathon Competitor & Finalist:** Built and presented rapid full-stack and backend architectural prototypes under intense 36-hour sprint constraints.
- Java Foundations Professional Certificate:** JetBrains & LinkedIn Learning (Core Java, Collections, Multithreading, OOP).
- Spring Boot, Spring & Hibernate for Beginners:** Udemy Verified Certification (REST APIs, Dependency Injection, JPA, ORM).
- REST APIs & Azure Cloud Fundamentals:** LinkedIn Learning.
- AWS Academy Graduate:** Cloud & Generative AI Foundations.
- Git, GitHub & SQL Certified:** Simplilearn.

EDUCATION

B.Tech in Computer Science & Engineering - SR Institute of Management and Technology, Lucknow

2023 - 2027

Senior Secondary (12th, Intermediate) - Indian Educational Inter College

2023 (52.40%)

High School (10th) - Indian Educational Inter College

2021 (75.33%)